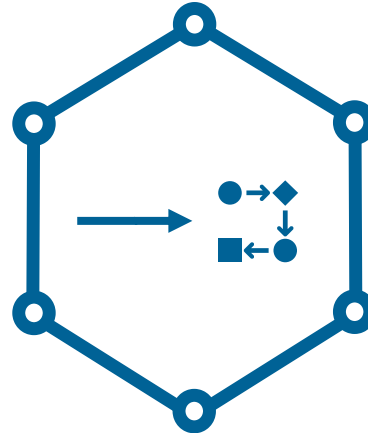




Knowledge Graph Creation Schema Design

Bonus

Emanuel Sallinger

Disclaimer:

- **Primer on how to select your nodes, edges, etc.**
- **Not on normalization, performance, ...**
- **Not on the constraints in your schema**
- **Not on knowledge representation, ML representations, etc. that may be part of your schema**



Slides courtesy of Vinay K. Chaudhri
University of Stanford

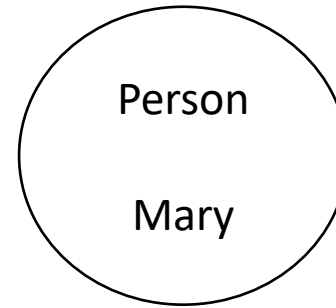
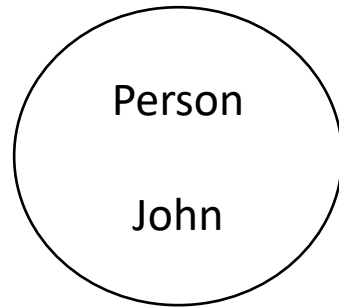
Even though there are some guidelines to design, there are also equivalent good choices.

Where else to look?

- **Relational Database Design**
- **RDF Design**
- **Property Graph Design**
- **Graph Database Design**
- ...

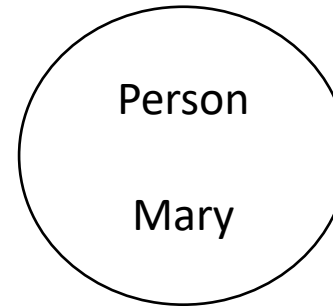
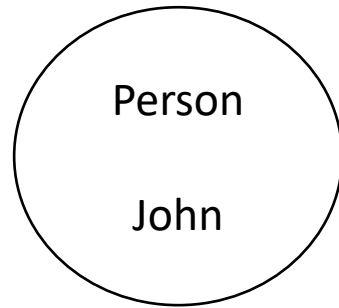
Choosing Nodes, Labels and Properties

- Nodes usually represent entities in a domain



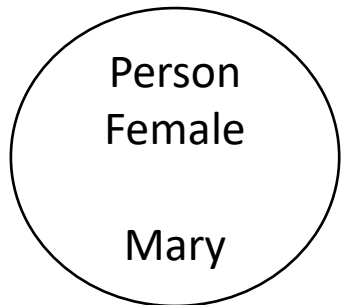
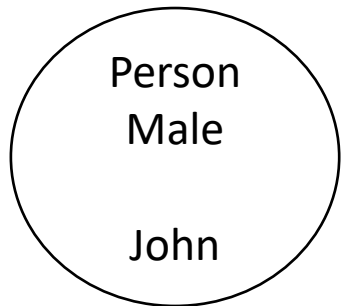
Choosing Nodes, Labels and Properties

- Nodes usually represent entities in a domain
 - How to represent gender?



Choosing Nodes, Labels and Properties

- Nodes usually represent entities in a domain
 - How to represent gender?



Introduce a new Label

Labels are like classes

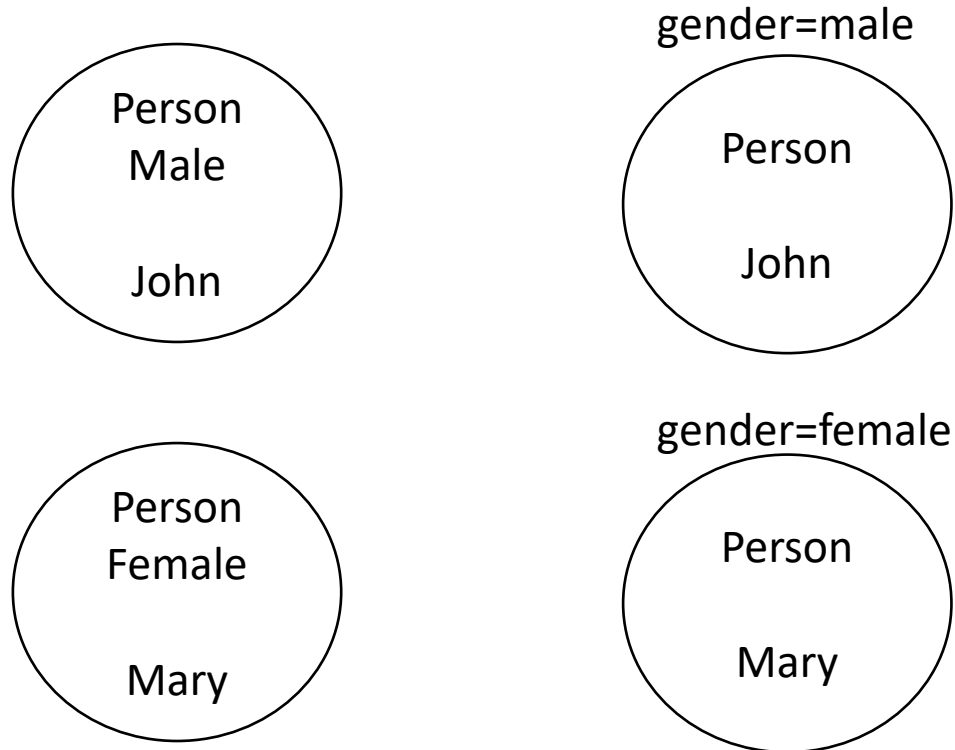
Labels should be natural

Labels should not change with time

Could benefit from indexing

Choosing Nodes, Labels and Properties

- Nodes usually represent entities in a domain
 - How to represent gender?

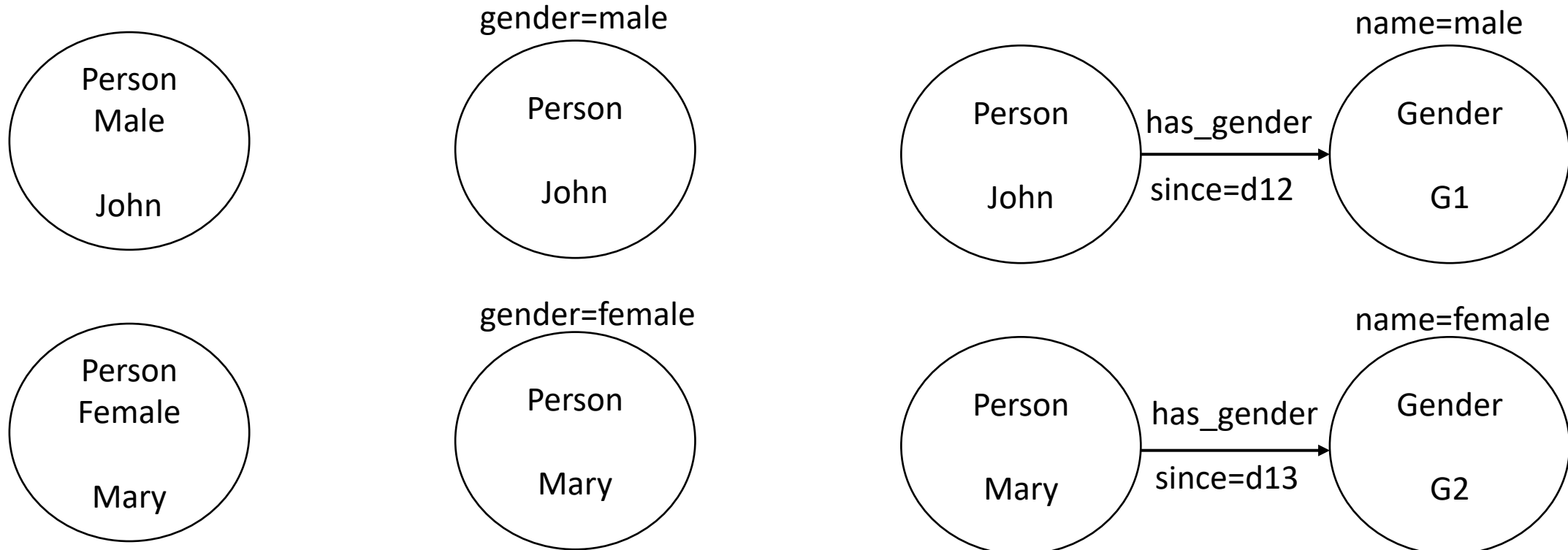


Introduce a new node property

Property should not change with time

Choosing Nodes, Labels and Properties

- Nodes usually represent entities in a domain
 - How to represent gender?



Properties

Addon

Person

ID	Name	Gender
1	John	Male
2	Mary	Female

Properties

Person

ID	Name	Gender
1	John	Male
2	Mary	Female

Labels

Person

ID	Name
1	John
2	Mary

Male

ID
1

Female

ID
2

Properties

Person

ID	Name	Gender
1	John	Male
2	Mary	Female

Labels

Person

ID	Name
1	John
2	Mary

Male

ID
1

Female

ID
2

Relationships

Addon

Person

ID	Name
1	John
2	Mary

Gender

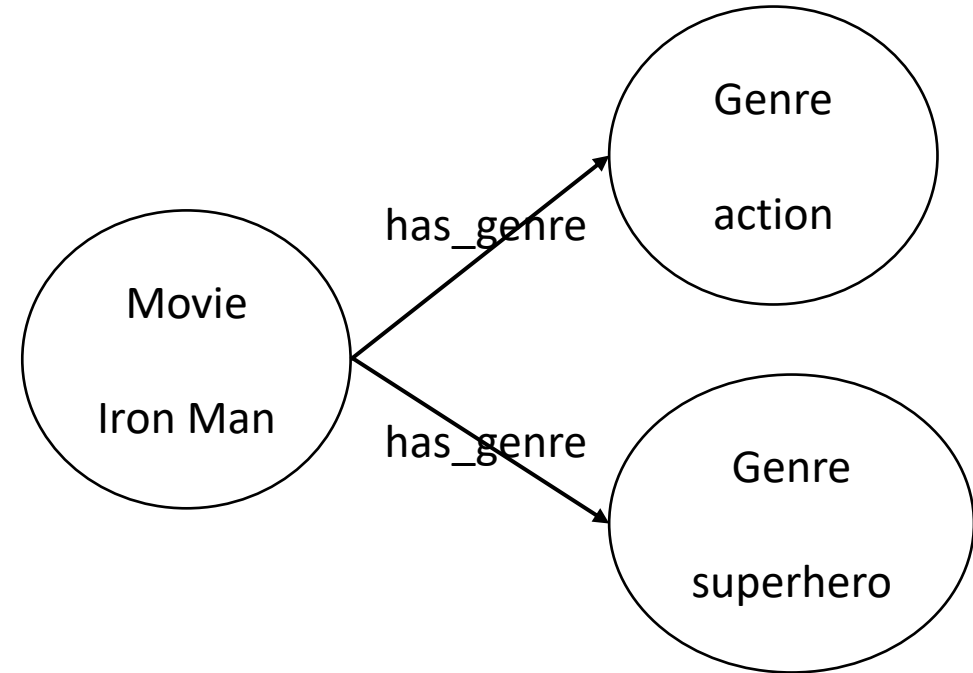
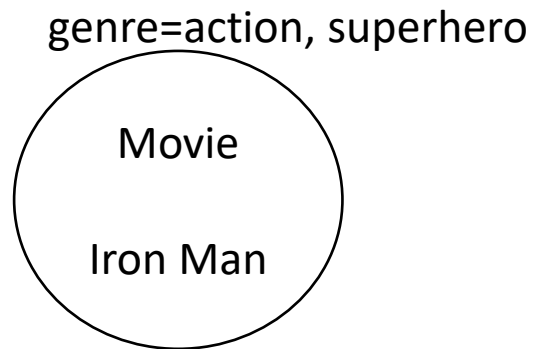
ID	Name
G1	Male
G2	Female

hasGender

ID	Gender	Since
1	G1	d12
2	G2	d13

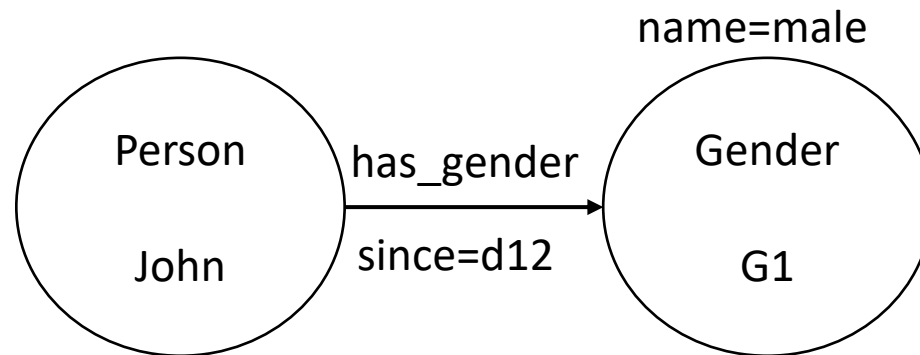
When to introduce a Relationship?

- When efficient access is required



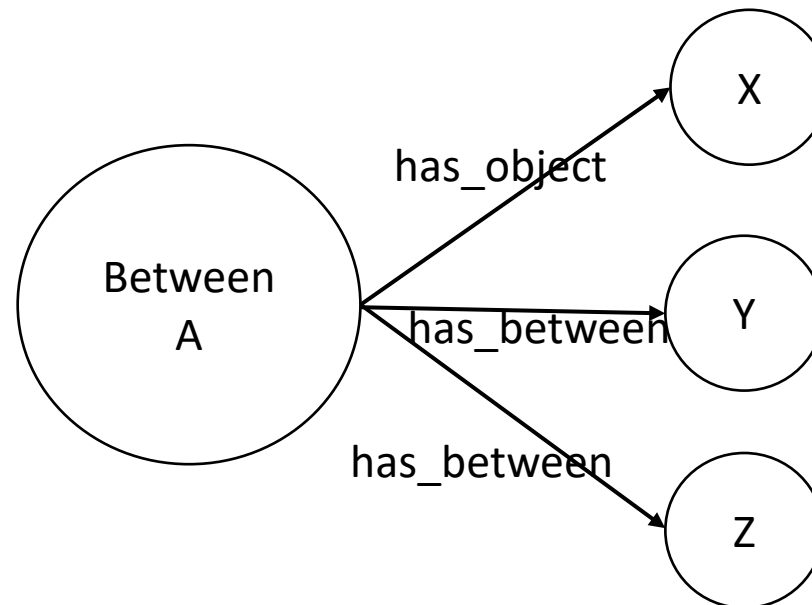
When to Introduce a Relationship Property

- Common use cases for relationship properties
 - Time varying relationships
 - Provenance
 - Confidence



What if your chosen representation cannot represent multi-ary relationships?

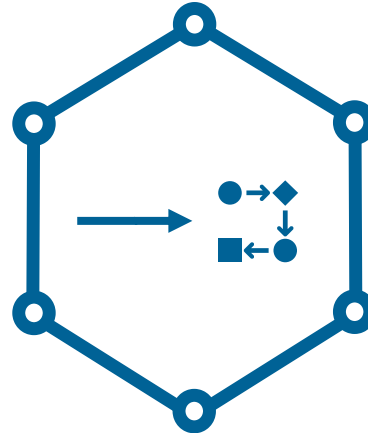
- Reification is a common technique to handle relationships with arity higher than 2
 - Create an object representing the relationship
 - Create objects for each argument of the relationship
 - Introduce relationships to connect them



X is between Y and Z

Where else to look?

- **Relational Database Design**
- **RDF Design**
- **Property Graph Design**
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- ...



Knowledge Graph Creation Schema Design

Bonus

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